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Application of the TRIZ creativity enhancement approach to the design of a passively adaptive compliant robotic gripper

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Abstract

Purpose – The essence of the conceptual design is getting the innovative projects or ideas to ensure the products with best performance. It has been proved that the theory of inventive problem solution (TRIZ) is a systematic methodology for innovation. The purpose of this paper is to illustrate the design of an adaptive robotic gripper as an engineering example to show the significance and approaches of applying TRIZ in getting the creative conceptual design ideas.

Design/methodology/approach – Gripping and holding of objects are key tasks for robotic manipulators. The development of universal grippers able to pick up unfamiliar objects of widely varying shapes and surfaces is a very challenging task. The requirement for new adaptive grippers is the ability to detect and recognize objects in their environments.

Findings – The main aim of this work is to show a systematic methodology for innovation as an effective procedure to enhance the capability of developing innovative products and to overcome the main design problems. The TRIZ method will be utilized in order to eliminate the technical contradictions which appear in the passively adaptive compliant robotic gripper.

Originality/value – The design of an adaptive robotic gripper as an engineering example is illustrated in this paper to show the significance and approaches of applying TRIZ in getting the creative conceptual design ideas.

Keywords Inventive principles, TRIZ method, Technical contradictions, Adaptive robotic gripper, Underactuation, Design, Robotics

Paper type Research paper

Introduction

The ability to solve inventive problems is at the core of the innovation process; however, the standard procedure to deal with them is to utilize random trial and error, despite the existence of several theories and methods. Many methodologies have been proposed to help process engineers accelerate complex design and propose creative ideas (Duran-Novoa *et al.*, 2011; Cortes Robles *et al.*, 2009).

In 1946, Genrich Altshuller started a statistical analysis of more than two million worldwide patents (Tong *et al.*, 2006); using that data, he recognized the principles on which the technical system developments and solutions are based. Since then, an algorithm for solving inventive problems has been perfected, and it is now available: this is the effort to advance the theory of inventive problem solution (TRIZ) (Altshuller and Shapiro, 1956; Fey and Rivin, 2005; Jovanović *et al.*, 2009) for all kinds of technical

problems as a scientific theory. According to TRIZ, a similar analogy can be defined for inventive problems: a model of problem is interfaced with a model of its solution by a set of tools.

TRIZ is an algorithmic approach for solving technical and technological problems (Justel *et al.*, 2006). Along with innovation, it brings efficiency into the process in that the suggested TRIZ principles provide definite guidelines for the engineers to think. However, TRIZ is a stronger conceptual than analytical tool. Although TRIZ can be used as a problem definition tool, its greatest strength lies in resolving contradictions (Stratton and Mann, 2003; Kim *et al.*, 2009; Yang and Chen, 2011) and solving problems defined using other techniques.

One of the most popular TRIZ tools is the “40 Inventive Principles” which consists of a group of deducted generic solutions of technical contradictions across many fields (Filippi *et al.*, 2011; Berdonosov and Redkolis, 2011; Nakagawa, 2011). Altshuller matrix organizes the principles depending on the contradiction they solve, which makes for an easy handling and positioning of the inherent contradiction in the forefront of the problem definition (Maarten Bonnema, 2011; Shirwaiker and Okudan, 2011; Ogot, 2011).

This study deals with the elimination of the contradictions that appear in adaptive robotic grippers using inventive principles recommended by the TRIZ method. Significant efforts have been made to find designs of adaptive robotic grippers simple enough to

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be easily built and controlled, in order to obtain practical systems. To overcome the limited success of the early designs, due to the cost of the control architecture, a special emphasis has been placed on the reduction of the number of degrees of freedom, thereby decreasing the number of actuators. The strategy for reducing the number of actuators while keeping the hand capability to adapt its shape to grasped object is referred to as underactuation. Papers Carrozza *et al.* (2004), Montambault and Gosselin (2001), Fukaya *et al.* (2000), Birglen (2011) and Daniel *et al.* (2011) show that underactuation allows the reproduction of most of the grasping behaviours of the human hand, without augmenting the mechanical and control complexity. Due to the multiple degrees of freedom and passive behavior, any compliant mechanism (Lu and Kota, 2002, 2003a, b, 2005; Kota *et al.*, 2003; Birglen and Gosselin, 2005; Luo *et al.*, 2005) can be considered as an underactuated mechanism, i.e. with fewer actuators than degrees of mobility. Finger compliance allows the gripper to passively conform to a wide range of objects while minimizing contact forces.

An adaptive (robotic) mechanical system is an adaptive system in which the ability to adapt to new external situations relies strictly on mechanical properties. In other words, an adaptive mechanical system is one in which some form of intelligence is embedded into the mechanics. In such a system, no sensors or complex controllers are required to perform the main task since the mechanical system itself will provide the required adaptive behavior. The main task of this study is to investigate the application of new inventive principles in adaptive underactuated robotic grippers to ensure adaptation to any irregular object.

TRIZ method

TRIZ is the Russian acronym for the Theory of Inventive Problem Solving developed by Genrich Altshuller in Russia in 1965. It is a systematic approach to creativity. Based on his analysis of 40,000 patents, Altshuller recognized that the same fundamental problems (or contradictions) in one area had been addressed by many inventions in other technological areas. He also found that the same fundamental solutions had been used over and over again. Based upon the 40,000 patents collected, Altshuller summarized 1,201 standard engineering problems, named contradictions, and 40 fundamental solutions to these problems, named Inventive Principles. In contrast to traditional approaches to creativity, the inventors using TRIZ are not only interested in searching for inventions in related fields, but also analogous problems in other fields that have previously solved the same contradiction.

TRIZ is a problem solving method that increases the ability to solve creative problems thanks to its capacity to give access to the best practices in all the technical domains. The main findings of TRIZ are:

- There are the trends in the evolution of technologies.
- All innovations start with application of a small number of inventive principles.
- The innovative solution of the problem removes the conflicts existing in the system.

During the patents analysis, Altshuller noted that technical contradictions could be expressed in terms of conflict between two parameters: one improved and the other one damaged. Only 39 parameters (Table I) were extracted to describe all the contradictions encountered in patents. Presenting a

Table I 39 engineering parameters of TRIZ

1	Weight of moving object
2	Weight of stationary
3	Length of moving object
4	Length of stationary
5	Area of moving object
6	Area of stationary
7	Volume of moving object
8	Volume of stationary
9	Speed
10	Force (intensity)
11	Stress of pressure
12	Shape
13	Stability of the object
14	Strength
15	Durability of moving object
16	Durability of non-moving object
17	Temperature
18	Illumination intensity
19	Use of energy by moving
20	Use of energy by stationary
21	Power
22	Loss of energy
23	Loss of substance
24	Loss of information
25	Loss of time
26	Quantity of substance
27	Reliability
28	Measurement accuracy
29	Manufacturing precision
30	Object-affected harmful
31	Harmful side effect
32	Ease of manufacture
33	Ease of operation
34	Ease of repair
35	Adaptability of versatility
36	Complexity of device
37	Complexity of control
38	Level of automation
39	Productivity

technical contradiction as a combination of two parameters requires a broad interpretation of them, so they are generic for any engineering field. Parameters and principles are put together in a tool where the contradiction formulation (problem formulation) and ways to solve it (principles) are present. Finally, they build a 39×39 matrix.

Resolving the passive compliant gripper design contradictions by TRIZ

Contradiction problems often occur in engineering. A technical contradiction arises when efforts to improve one system characteristic degrades another one. For instance, adding more steel to increase the strength of a car makes a car heavier. In TRIZ, a technical contradiction can be solved by using the contradiction matrix in the design process. A technical conflict problem can be solved in four steps by using the contradiction matrix table:

- *Step 1.* Analyze the attributes of the problem and determine the parameter “feature to improve” and “undesired result”.
- *Step 2.* Identify both parameters among the 39 (Table I).
- *Step 3.* Use the contradiction matrix.
- *Step 4.* Attempt to resolve the conflict problem by using three or four recommend inventive principles that are listed in the intersecting cell. The inventive principles are used to facilitate the development of useful “concepts of solution” for innovative situations. During the step, if the principles proposed in the cell of the contradiction matrix do not generate a solution, the user can try to use the other ones.

The general problem solving process by TRIZ is shown in Figure 1 where the essence of the TRIZ philosophy is captured.

To resolve the contradiction, from Table II, the used principles for the passive compliant gripper design are 15, 3, 35 and 6:

15. Dynamics

Allow (or design) the characteristics of an object, external environment, or process to change to be optimal or to find an optimal operating condition.

Divide an object into parts capable of movement relative to each other.

If an object (or process) is rigid or inflexible, make it movable or adaptive.

Analyzing the inventive principle leads to the solution that the robotic gripper should be designed to always have optimal characteristics during operation. It means that the gripper structure does not have concentrated stress but that it is distributed through the whole structure. It leads to the solution that the gripper should be made as a fully compliant mechanism

with distributed compliance (Figure 2). This is proven by the last sentence in the inventive principle which emphasizes that the gripper should be movable or adaptive.

According to the finite element simulations, it was proven that the gripper could accommodate different shapes and sizes of the grasping objects. Figure 2 shows the verification of the gripper behavior for seven different shapes and sizes of grasping objects. During the simulations the grasping objects were fixed.

Afterwards, the verification of the gripper prototype was conducted as shown in Figure 3. The radius of the used cylindrical grasping object was $r = 25$ mm. The gripper thickness was 10 mm and it was not changed during experimental testing. The gripper dimension is scalable so it can be easily applied to a wide range of grasping objects, i.e. macro and micro domain.

The two-fingered gripper structure can be useful in experimental analysis of the finger characteristics, even though, two-fingered grippers can show unstable behavior during grasping and holding of objects for real grasping applications in robotics. For such a purpose, it is useful to model multi-fingered grippers with three or more fingers. Figure 4(a) shows three-fingered gripper accommodation to a cylindrical object (convex grasping shape). Figure 4(b) shows four-fingered gripper accommodation to a cylindrical grasping object. In both cases, the single actuation principle was used. It means that with one actuator and without any control procedure the gripper should provide safe and full accommodation to the object. This is the main advantage of the passive mechanisms with underactuation. Figure 4(c) shows

Figure 1 The generic problem solving process by TRIZ

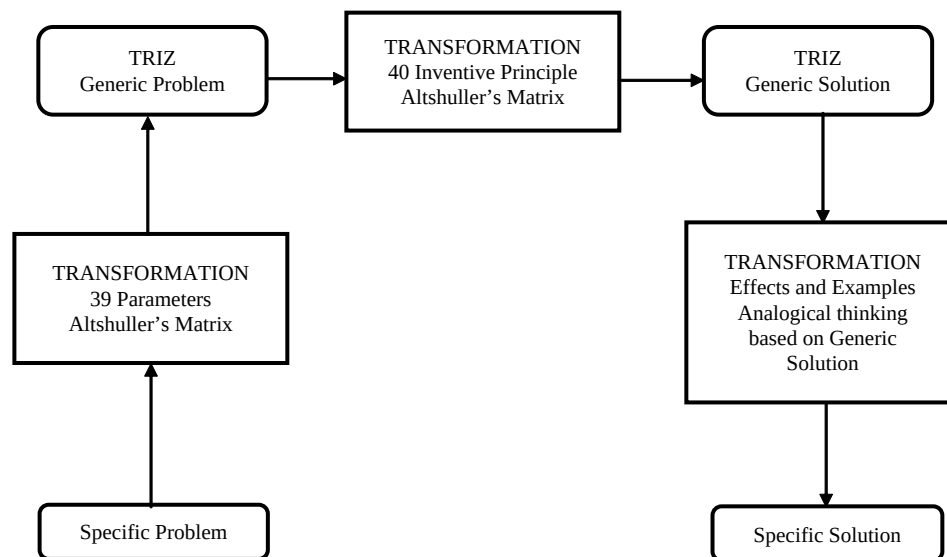
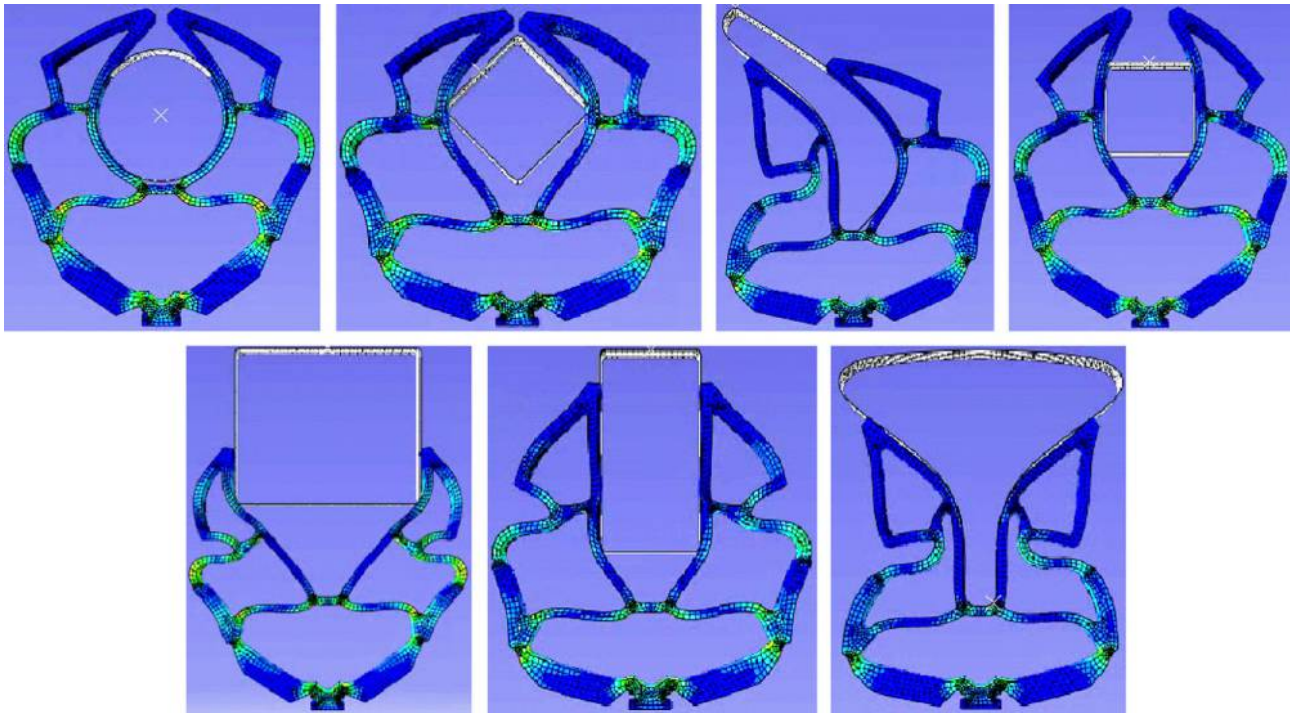


Table II Section of the used contradiction matrix

Feature to improve	33. Ease of operation	Undesired result (degraded feature) 36. Complexity of device	37. Complexity of control
35. Adaptability of versatility		15, 28, 29, 37	
38. Level of automation	1, 3, 12, 34		
27. Reliability		1, 13, 35	
32. Ease of manufacture			1, 6, 11, 28

Figure 2 Verification of the gripper functions for different shapes of grasping objects**Figure 3** Verification of the gripper prototype for convex grasping shape

the four-fingered gripper behavior for the concave grasping shape of the object. The full adaptation of all fingers to the concave grasping shape can be also noticed:

3. Local quality

Change an object's structure from uniform to non-uniform, change an external environment (or external influence) from uniform to non-uniform.

Make each part of an object function in conditions most suitable for its operation.

Make each part of an object fulfill a different and useful function.

Analyzing the inventive principle leads to the solution that the robotic gripper should have a non-uniform structure. It leads to the solution that the gripper could be modeled with embedded sensors as parts of its structure (Figure 3). The sensors material should have the same elastic features as the rest of the gripper structure. One of the possible solutions is to use silicone rubber for the main structure and conductive silicone rubber for the embedded sensors design. Further analysis of the principle leads to each part (sensor) having the most suitable conditions (place) for its operation.

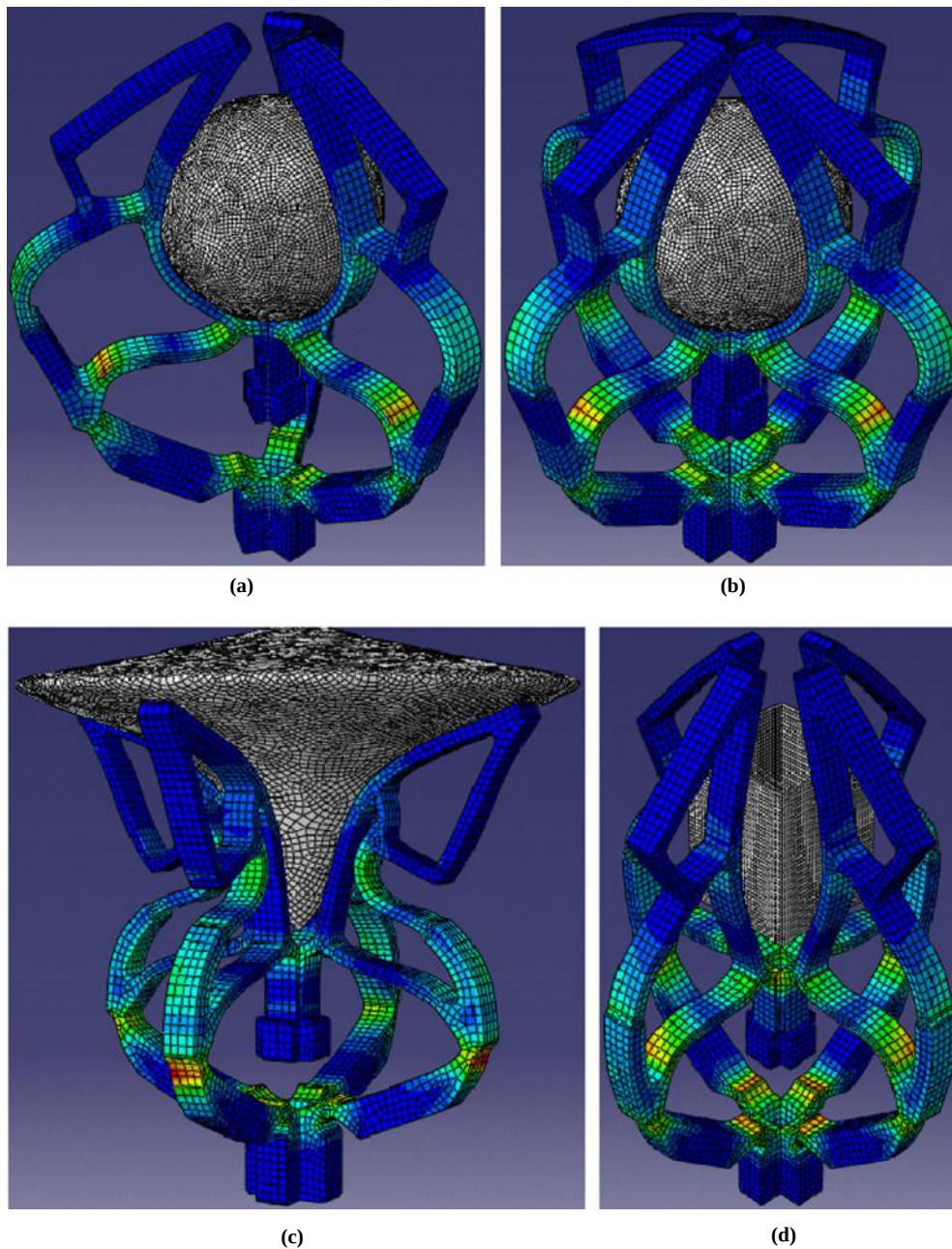
Figure 5 shows the main gripper features. These are:

- The whole gripper structure represents one passive elastic structure (Figure 5(a)).
- For one active input, the gripper has multi-output contact points (Figure 5(b)).
- One active input actuator (Figure 5(b)).
- The sensors of conductive silicone rubber embedded in the gripper structure (Figure 5(a)).

A control algorithm based on the changes in embedded sensors voltage could be derived to perform tasks of detection and recognition of a grasping object. Simultaneously, the controller could provide the input displacement signal according to the shape of the fixed grasping object. Figure 6 shows one gripper with an embedded sensor. The finger input displacement is also depicted.

It is important to mention that the grasping objects were fixed during the measurement procedure. The embedded sensor has different deformations for different grasping patterns. Figure 7 shows some examples of grasping objects and correspondent deformations of the embedded sensor. Here, segment t_3 was used as grasping shape indicator.

The embedded sensor could be used to detect and recognize grasping objects according to its deformations. Based on experimental tests, three main grasping patterns could be

Figure 4 Multi-fingered grippers with: (a) three fingers and (b)-(d) four fingers

created as shown in Figure 8. These are “concave”, “convex” and “no object” patterns. Each of these grasping patterns requires different input displacements for the gripper. Measured changes in normalized voltage of the embedded sensors are shown in Figure 7 for the three grasping patterns:

35. Parameter changes

- Change an object's physical state (e.g. to a gas, liquid, or solid).
- Change the concentration or consistency.
- Change the degree of flexibility.
- Change the temperature.

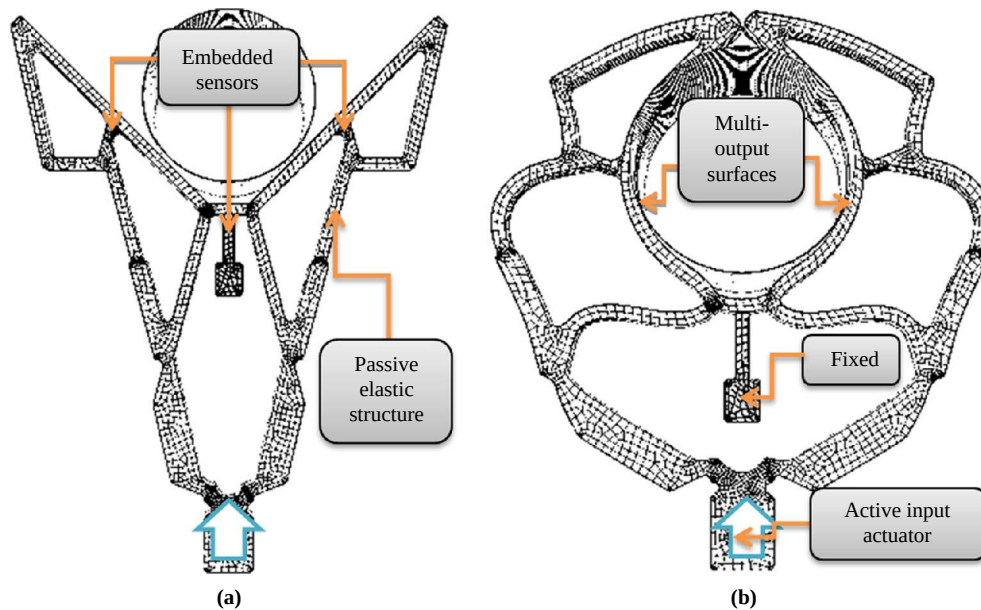
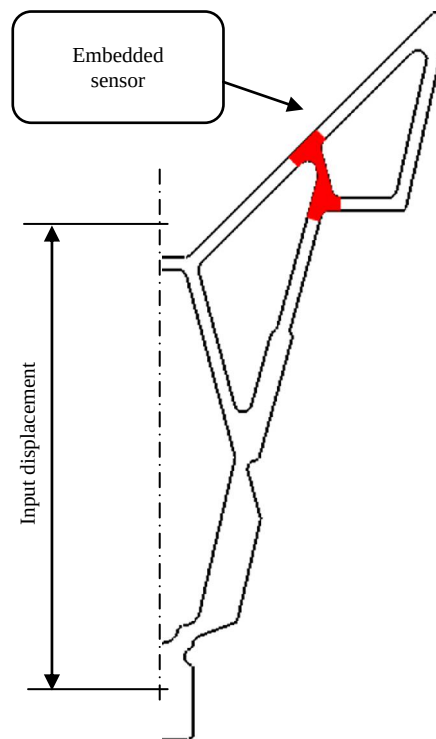
Analyzing the inventive principle leads to the solution that different types of the silicone rubber and conductive silicone

rubber should be tested to determine which stiffness is the best for the gripper application.

Figure 9(a) shows the characteristic of the used conductive silicone rubber during stop-time under compression. Several elements were compressed at 25 per cent and kept thus deformed for two minutes before being unloaded; this was repeated five times. The load and unload velocity was 30 mm/min. Also, during the stop-time, there was a very small decrease of the sensor-element resistance, as shown in Figure 9(b). Therefore, it can be stated that conductive silicone rubber has good strength and elastic quality:

6. Universality

- Make a part or object perform multiple functions; eliminate the need for other parts.

Figure 5 The main features of the adaptive gripper**Figure 6** Finger design with embedded sensor

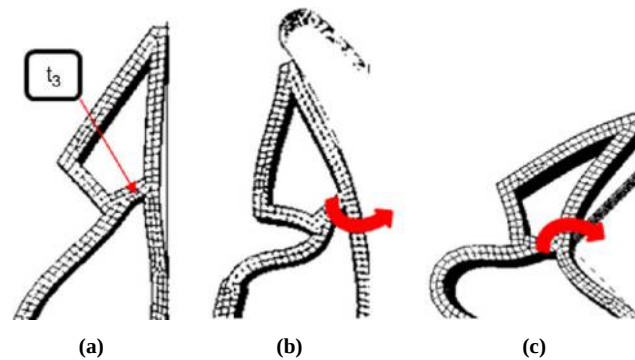
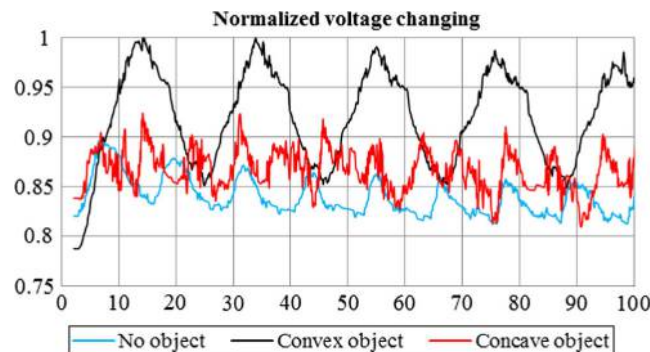
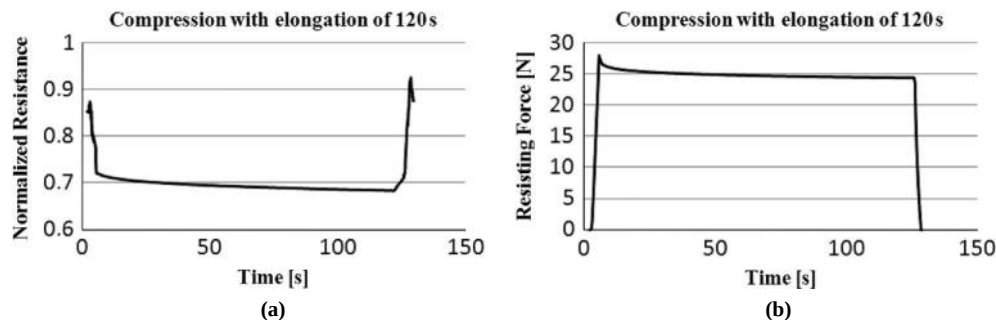
Analyzing the inventive principle leads to the solution that some parts (embedded sensors) of the gripper structure should perform multiple functions, i.e. measuring and accommodation at the same time (Figure 5).

Gripper manufacturing

The gripper manufacturing process was presented in Issa *et al.* (2011). The moulding tool for the gripper manufacturing is shown in Figure 10. The process follows the next steps:

- 1 The tool cavities are filled with raw silicone rubber.
- 2 Press-curing at $T = 180^{\circ}\text{C}$ and pressure of $p = 26 \text{ kN}$ for about 30 min.

The extracted gripper from the tool is shown as well. In the future work, the sensors will be made of conductive silicone rubber which will be more elastic than silicone rubber for the gripper structure to ensure more deformation in the sensors. This paper explains the production of the gripper structure without sensors. At first, silicone rubber with different shore hardness was used,

Figure 7 Different grasping objects grouped in three main grasping patterns: (a) no object, (b) concave shape and (c) convex shape**Figure 8** Changes in embedded sensor voltage for three grasping patterns**Figure 9** Sensor resistance changing for compression test with stop-time two minutes while compression of 25 percent

Notes: (a) Time-resistance characteristic; (b) time-resisting force characteristic

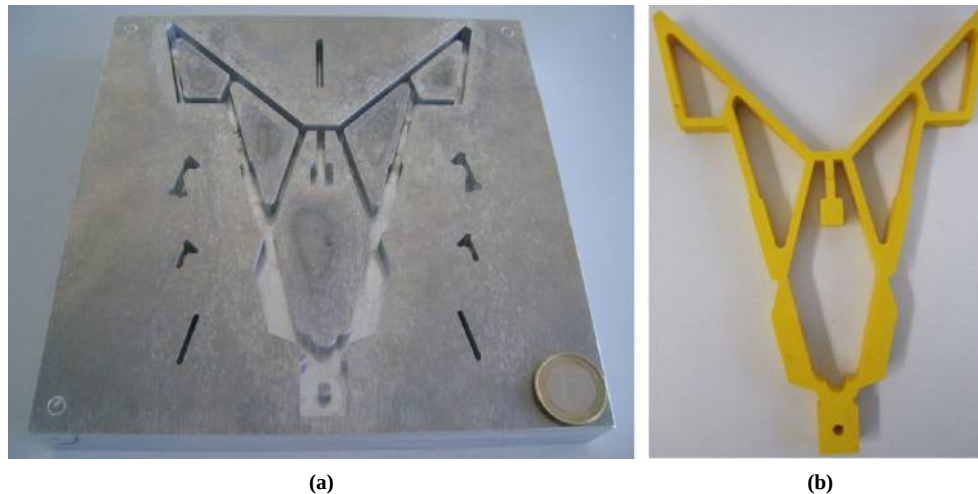
i.e. 60, 70 and 80. According to the experimental test, the best silicone shore was 70, therefore, Elastosil R420/70 was used.

Conclusion

TRIZ is a problem solving method that increases the ability to solve creative problems thanks to its capacity to give access to the best practices in all the technical domains. This method is used to assist in the step-change innovation. It can help the engineers to get an innovative conceptual idea for the design project by following the innovation roadmap with some tool kits provided by TRIZ. TRIZ is a treasure which includes the correct methodology and related knowledge base.

The research described in this paper demonstrates that the development of innovative design for a passive compliant robotic gripper with embedded sensors was achieved within the framework of TRIZ methodology. Four steps of evolution of the robotic gripper design were achieved by contradiction analyses in TRIZ. The final design of the passive compliant robotic gripper with embedded sensors was derived from the four inventive principles of TRIZ in the following order:

- 1 dynamics;
- 2 local quality;
- 3 parameter changes; and
- 4 universality.

Figure 10 The moulding tool for gripper manufacturing and one extracted gripper without embedded sensors

According to these inventive principles a new gripper principle was utilized here. There are certain passively adaptive compliant gripper designs, but their applications are only in the micro domain. For example, article Bruzzone and Bozzini (2011) described gripper design with passive adaptive feature but its limitation lay in concentrated stress in the compliant joints and limited displacement range of the joints. Another design of passively adaptive gripper was introduced in Dechev *et al.* (2004) but its application was also in the micro domain, and full accommodation of the gripper to any grasping object was not achieved. So far, there have been no investigations on the application of passively underactuated compliant grippers in the macro domain but only in the micro or nano domain (Zubir and Shirinzadeh, 2008; Liaw *et al.*, 2008). The main advantages of the utilized gripper lie in its distributed compliance, simple manufacturing process, low cost, easy adaptation to any irregular object in the wide range of size, and implementation of embedded sensors for better sensing capability. It is possible to scale gripper dimensions according to its application.

The experimental prototype of the proposed robotic compliant gripper was designed in the Laboratory Mechanism Technology, Faculty of Mechanical Engineering at Ilmenau University of Technology, Germany.

TRIZ is a logical, knowledge-based methodology for early stages of the design process. The application of traditional TRIZ accelerates the search for breakthrough solutions and gives users the ability to reach greater levels of product performance. The results of the research carried out were realized in the development of a new methodology for passive compliance in the robotic gripper with embedded sensors, by adapting one of the most useful tools of TRIZ – the inventive principles. The results of the proposed methodology of TRIZ were also demonstrated by the case study.

On the rapidity point of view, the TRIZ theory has a drawback due to its level of application of knowledge. However, with TRIZ designers propose solutions more quickly than with other methods focused on creativity stimulation and innovation. Moreover, TRIZ places designers under good conditions by giving them some directions to express their creativity.

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